

Summer school ISIMA - OTH

Phase 2

Mathematical Modeling and Optimization with Julia

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Lesson 3 - Modelling Tricks with Integer Variables

August 2026

Logic, activation, disjunctions and linearization

Represent a yes–no decision

Let $y \in \{0, 1\}$ represent a statement or a discrete decision:

$y = 1 \iff$ the decision is selected or the statement is true.

Typical meanings:

- ▶ open a facility;
- ▶ use a machine;
- ▶ select a project;
- ▶ satisfy a condition;
- ▶ choose one branch of a disjunction.

The modelling principle:

Translate the truth table or the intended operating modes into linear inequalities.

Bounds on continuous variables are often essential.

Example: represent warehouse-opening decisions

A company may open three warehouses. Define

$$y_j = \begin{cases} 1, & \text{if warehouse } j \text{ is opened,} \\ 0, & \text{otherwise,} \end{cases} \quad j = 1, 2, 3.$$

Then common requirements become linear constraints:

$$y_1 + y_2 + y_3 \leq 2 \quad \text{open at most two warehouses,}$$

$$y_1 \leq y_2 \quad \text{opening 1 requires opening 2,}$$

$$y_2 + y_3 \geq 1 \quad \text{open warehouse 2 or 3.}$$

Model a cost paid only when an activity starts

Suppose producing $x \geq 0$ costs cx plus a fixed charge F whenever $x > 0$.

$$\min cx + Fy \quad 0 \leq x \leq Uy, \quad y \in \{0, 1\}.$$

If $y = 0$, then $x = 0$: production is closed.

If $y = 1$, then $0 \leq x \leq U$: production is allowed.

U must be a valid upper bound on x . A tight U gives a stronger LP relaxation.

Example: pay a machine setup cost

A machine produces x units. Starting it costs 100 euros and each unit costs 3 euros. Its capacity is 40 units.

$$\min 3x + 100y, \quad 0 \leq x \leq 40y, \quad y \in \{0, 1\}.$$

- ▶ If $y = 0$, then $x = 0$ and no setup cost is paid.
- ▶ If $x > 0$, then $y = 1$ and the 100-euro charge is paid.

If at least 10 units must be produced after setup, replace the linking constraint by

$$10y \leq x \leq 40y.$$

Model an activity that is either off or bounded

A machine is either idle or operates between L and U units:

$$x \in \{0\} \cup [L, U].$$

x is called a semicontinuous variable.

An exact MILP formulation is

$$Ly \leq x \leq Uy, \quad y \in \{0, 1\}.$$

- ▶ $y = 0$ forces $x = 0$.
- ▶ $y = 1$ forces $L \leq x \leq U$.

Example: operate a bounded generator

A generator is either stopped or produces between 20 and 100 MW:

$$x \in \{0\} \cup [20, 100].$$

Introduce $y \in \{0, 1\}$ and impose

$$20y \leq x \leq 100y.$$

y	Resulting domain of x	Operating state
0	$x = 0$	stopped
1	$20 \leq x \leq 100$	operating

Enforce a constraint only if a decision is selected

To model

$$y = 1 \implies a^\top x \leq b,$$

use the big- M inequality

$$a^\top x \leq b + M(1 - y).$$

- ▶ If $y = 1$, the original constraint is enforced.
- ▶ If $y = 0$, it is relaxed by M .

A valid value is $M \geq \max\{a^\top x - b : x \in X\}$, where X contains the other known bounds and constraints.

Example: enforce a contract capacity limit

Let $0 \leq x \leq 10$. If a particular contract is selected ($y = 1$), production must satisfy

$$x \leq 4.$$

Because the largest possible violation is $10 - 4 = 6$, choose $M = 6$:

$$x \leq 4 + 6(1 - y).$$

- ▶ If $y = 1$: $x \leq 4$.
- ▶ If $y = 0$: $x \leq 10$, so the additional limit disappears.

Identify whether a condition holds

To model

$$y = 1 \iff a^\top x \leq b,$$

one must encode both directions.

With a tolerance or discrete gap $\varepsilon > 0$:

$$\begin{aligned} a^\top x &\leq b + M^+(1 - y), \\ a^\top x &\geq b + \varepsilon - M^-y. \end{aligned}$$

Let

$$U_a = \max_{x \in X} a^\top x, \quad L_a = \min_{x \in X} a^\top x.$$

Tight deactivation constants are

$$M^+ = \max\{0, U_a - b\}, \quad M^- = \max\{0, b + \varepsilon - L_a\}.$$

M^+ relaxes the upper bound when $y = 0$; M^- relaxes the lower bound when $y = 1$.

The gap ε must be meaningful for the application.

Example: identify whether a threshold is respected

Suppose x is integer, $0 \leq x \leq 10$, and

$$y = 1 \iff x \leq 4.$$

Because x is integer, take $\varepsilon = 1$. Then

$$L_a = 0, \quad U_a = 10, \quad M^+ = 10 - 4 = 6, \quad M^- = 4 + 1 - 0 = 5.$$

The formulation is

$$x \leq 4 + 6(1 - y), \quad x \geq 5 - 5y.$$

If $y = 1$, then $x \leq 4$; if $y = 0$, then $x \geq 5$.

Identify when all conditions are satisfied

Let $z = 1$ if and only if $x_1 = x_2 = 1$:

$$z = x_1 \wedge x_2.$$

The linear formulation is

$$z \leq x_1, \quad z \leq x_2, \quad z \geq x_1 + x_2 - 1, \quad z \in \{0, 1\}.$$

For n binary variables,

$$z \leq x_i \quad (i = 1, \dots, n), \quad z \geq \sum_{i=1}^n x_i - (n - 1).$$

Example: earn a bonus for selecting both projects

Projects 1 and 2 generate an additional profit of 20 only when both are selected. Let $x_1, x_2 \in \{0, 1\}$ and introduce $z = x_1 \wedge x_2$:

$$z \leq x_1, \quad z \leq x_2, \quad z \geq x_1 + x_2 - 1.$$

Add $20z$ to the objective. The possible values are

(x_1, x_2)	$(0, 0)$	$(0, 1)$	$(1, 0)$	$(1, 1)$
z	0	0	0	1

Thus the bonus is obtained only in the final case.

Identify when at least one condition is satisfied

Let $z = 1$ if at least one of x_1, x_2 equals one:

$$z = x_1 \vee x_2.$$

The linear formulation is

$$z \geq x_1, \quad z \geq x_2, \quad z \leq x_1 + x_2, \quad z \in \{0, 1\}.$$

For n binary variables,

$$z \geq x_i \quad (i = 1, \dots, n), \quad z \leq \sum_{i=1}^n x_i.$$

Example: detect available backup equipment

Let x_1 and x_2 indicate whether two backup generators are installed. Define $z = 1$ when at least one generator is available:

$$z \geq x_1, \quad z \geq x_2, \quad z \leq x_1 + x_2.$$

(x_1, x_2)	$(0, 0)$	$(0, 1)$	$(1, 0)$	$(1, 1)$
z	0	1	1	1

A reliability requirement such as “backup must be available” is then simply $z = 1$.

Model opposite or mutually exclusive decisions

For binary variables x, y, z :

Logical relation	Meaning	Linear formulation
$z = \neg x$	opposite values	$z = 1 - x$
$x \Rightarrow y$	precedence	$x \leq y$
$x \Leftrightarrow y$	same value	$x = y$
$z = x \oplus y$	exactly one true	$z = x + y - 2w$

For XOR, $w = x \wedge y$ is binary and satisfies

$$w \leq x, \quad w \leq y, \quad w \geq x + y - 1.$$

Example: choose exactly one transport mode

Let $x_R = 1$ for road transport and $x_T = 1$ for rail transport. Exactly one mode must be selected:

$$x_R + x_T = 1.$$

This single equality is an XOR relation:

$$(x_R, x_T) \in \{(1, 0), (0, 1)\}.$$

If road transport also requires insurance, represented by y , add the implication

$$x_R \leq y.$$

If insurance is purchased exactly when road is chosen, write $x_R = y$.

Choose a prescribed number of alternatives

For binary variables x_i :

$$\text{at most } k : \sum_i x_i \leq k,$$

$$\text{at least } k : \sum_i x_i \geq k,$$

$$\text{exactly } k : \sum_i x_i = k.$$

These constraints are often combined with implications or incompatibilities.

Example: form a project portfolio

Five projects are available, with $x_i = 1$ if project i is selected.

Choose exactly three projects:

$$\sum_{i=1}^5 x_i = 3.$$

Projects 1 and 2 are incompatible:

$$x_1 + x_2 \leq 1.$$

If project 5 is selected, at least two of the other projects must be selected:

$$\sum_{i=1}^4 x_i \geq 2x_5.$$

Require at least one of two constraints to hold

Suppose at least one of the following must hold:

$$a^{\top}x \leq b \quad \text{or} \quad c^{\top}x \leq d.$$

Introduce $y \in \{0, 1\}$ and write

$$\begin{aligned} a^{\top}x &\leq b + M_1y, \\ c^{\top}x &\leq d + M_2(1 - y). \end{aligned}$$

If $y = 0$, the first constraint is active; if $y = 1$, the second is active.

This enforces at least one branch. Both branches may hold simultaneously.

Example: keep a variable outside a forbidden interval

Let $0 \leq x \leq 10$. We require

$$x \leq 2 \quad \text{or} \quad x \geq 8.$$

Introduce $y \in \{0, 1\}$:

$$x \leq 2 + 8y, \quad x \geq 8y.$$

- ▶ If $y = 0$, the first inequality gives $x \leq 2$.
- ▶ If $y = 1$, the second inequality gives $x \geq 8$.

Thus no feasible solution can satisfy $2 < x < 8$.

Apply a term only when two decisions are selected

To replace the nonlinear term

$$z = xy, \quad x, y \in \{0, 1\},$$

use

$$z \leq x, \quad z \leq y, \quad z \geq x + y - 1, \quad z \geq 0.$$

Here z may be declared continuous: the inequalities and binary values of x, y already force $z \in \{0, 1\}$.

Example: linearize a pairwise interaction cost

Opening facilities 1 and 2 simultaneously creates an additional coordination cost of 30:

$$30x_1x_2.$$

Introduce a continuous variable $z \geq 0$ and impose

$$z \leq x_1, \quad z \leq x_2, \quad z \geq x_1 + x_2 - 1.$$

Replace the nonlinear term by $30z$. Since $z = 1$ exactly when $x_1 = x_2 = 1$, the cost is charged in precisely that case.

Keep a quantity only when an option is selected

Let $y \in \{0, 1\}$, $L \leq x \leq U$, and $z = yx$. The exact linearization is

$$\begin{aligned} z &\geq Ly, & z &\leq Uy, \\ z &\geq x - U(1 - y), & z &\leq x - L(1 - y). \end{aligned}$$

- ▶ If $y = 0$, the first pair forces $z = 0$.
- ▶ If $y = 1$, the second pair forces $z = x$.

If $0 \leq x \leq U$, this simplifies to

$$0 \leq z \leq Uy, \quad z \leq x, \quad z \geq x - U(1 - y).$$

Example: count sales only when a promotion is active

Let $0 \leq x \leq 10$ be the number of units sold and $y = 1$ if a promotion is active. We need $z = yx$, the promoted sales.

Because $L = 0$ and $U = 10$, impose

$$0 \leq z \leq 10y, \quad z \leq x, \quad z \geq x - 10(1 - y).$$

For $x = 7$:

$$y = 0 \Rightarrow z = 0, \quad y = 1 \Rightarrow z = 7.$$

The nonlinear product does not appear in the MILP.

Measure an unsigned deviation

To minimize or upper-bound $|x|$, introduce $t \geq 0$:

$$t \geq x, \quad t \geq -x.$$

Then minimizing t forces $t = |x|$.

Without minimization or another upper pressure on t , these inequalities give only $t \geq |x|$, not equality. In this case, we will need extra binary variables.

Example: minimize deviation from a target

Let q be production and let the target be 100 units. We want to minimize

$$|q - 100|.$$

Introduce $t \geq 0$ and write

$$t \geq q - 100, \quad t \geq 100 - q, \quad \min t.$$

For example:

$$q = 92 \Rightarrow t = 8, \quad q = 105 \Rightarrow t = 5$$

at an optimal solution for the chosen value of q .

Represent the largest or smallest of several quantities

To model $z = \max\{x_1, \dots, x_n\}$ in a minimization context:

$$z \geq x_i \quad (i = 1, \dots, n).$$

Minimizing z gives equality automatically.

To enforce equality independently of the objective, we need to introduce binary variables y_i to represent the selection of x_i as the maximum.

$$z \geq x_i, \quad z \leq x_i + M_i(1 - y_i), \quad \sum_i y_i = 1.$$

Example: minimize the makespan

Two machines finish their work at times C_1 and C_2 . The makespan is

$$C_{\max} = \max\{C_1, C_2\}.$$

Introduce z and solve

$$\min z$$

subject to

$$z \geq C_1, \quad z \geq C_2.$$

If $C_1 = 7$ and $C_2 = 11$, the smallest feasible value is $z = 11$, exactly the completion time of the last machine.

Model a cost or response that changes by segments

For breakpoints $(\bar{x}_k, \bar{f}_k)$, a convex combination uses

$$x = \sum_k \lambda_k \bar{x}_k, \quad f = \sum_k \lambda_k \bar{f}_k, \quad \sum_k \lambda_k = 1, \quad \lambda_k \geq 0.$$

Additional adjacency restrictions are required for a general piecewise-linear graph:

- ▶ SOS2 constraints, if supported by the solver;
- ▶ binary segment-selection variables;
- ▶ incremental formulations.

Example: model shipping quantity discounts

The shipping cost has breakpoints

$$(\bar{x}_k, \bar{f}_k) \in \{(0, 0), (10, 50), (20, 80)\}.$$

Use

$$x = 10\lambda_2 + 20\lambda_3, \quad f = 50\lambda_2 + 80\lambda_3, \quad \lambda_1 + \lambda_2 + \lambda_3 = 1.$$

An SOS2 restriction permits positive weights only on adjacent breakpoints. For $x = 15$, interpolation between $(10, 50)$ and $(20, 80)$ gives

$$\lambda_2 = \lambda_3 = 0.5, \quad f = 65.$$

Build formulations that are both correct and efficient

Good practice

- ▶ derive tight variable bounds;
- ▶ compute a separate M for each constraint;
- ▶ use indicator or perspective formulations when appropriate;
- ▶ exploit solver-supported SOS and semicontinuous variables.

Common mistakes

- ▶ choosing an arbitrary huge M ;
- ▶ encoding only one direction of an equivalence;
- ▶ forgetting the role of the objective;
- ▶ attempting to model a strict inequality;
- ▶ omitting valid bounds on continuous variables.

Example: a correct big- M can still be unnecessarily weak

Consider

$$0 \leq x \leq 10, \quad y = 1 \Rightarrow x \leq 4.$$

The tight formulation uses $M = 6$:

$$x \leq 4 + 6(1 - y).$$

At the fractional LP point $y = 0.5$, this gives $x \leq 7$. Using $M = 10^6$ instead gives

$$x \leq 500004,$$

which adds no useful restriction beyond $x \leq 10$.

Both formulations are integer-valid, but the first provides a much stronger relaxation.

Choose a construction for each modelling need

Modelling need	Basic construction
Pay a fixed charge	$0 \leq x \leq Uy$ and add Fy
Activate a constraint	indicator, or $a^\top x \leq b + M(1 - y)$
AND / binary product	upper bounds plus $z \geq \sum x_i - (n - 1)$
OR	lower bounds plus $z \leq \sum x_i$
Choose one alternative	$\sum_i y_i = 1$
Binary $\times$ continuous	four bounded-product inequalities
Maximum in minimization	epigraph inequalities $z \geq x_i$
General disjunction	branch binaries plus tight deactivation bounds

The final check is always the same: fix the binary variables to every possible value and verify the resulting feasible set.